Numerical Linear Algebra
Lecture 1: Course Overview and Matrix-Vector Multiplication

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Outline

1 Course Overview

2 Motivation

3 Matrix-Vector Multiplication

Course Overview

Do you belong here?

- Prerequisite/Co-requisite:
  - This must NOT be your first course in linear algebra
  - Fundamentals of computing

Course Description

- What is numerical linear algebra?
  - Solving linear algebra problems using efficient algorithms on computers

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- Topics: Direct and iterative methods for solving simultaneous linear equations. Matrix factorization and decomposition. Conditioning, stability and efficiency. Computation of eigenvalues and eigenvectors
- Required Textbook: Numerical Linear Algebra by L. N. Trefethen and D. Bau
- Reference Books: Linear Algebra and its Applications by G. Strang/D. Lay

Course Policy

- Assignments (written and/or programming)
- Exams (written and/or take-home programming)
- Grading
  - Assignments :40%
  - Mid term exam: 30%
  - Final exam: 30%

Course Outline

- Matrix fundamentals, multiplications, orthogonality, norms and SVD (2.5 weeks)
- QR factorization, projectors, Gram-Schmidt algorithm, Householder triangulation (2 weeks)
- Conditioning and stability (1.5 weeks)
- Solution of linear system of equations: Gaussian elimination, pivoting, LU factorization, Cholesky factorization (2 weeks)
- Eigenvalue problems, Hasenberg tridiagonalization, Reyleigh quotient, power method, inverse power method and shifting (2.5 weeks)
- Iterative methods, Lanczos algorithm (1 week)
- Linear algebra software packages (1 week)

Python Labs

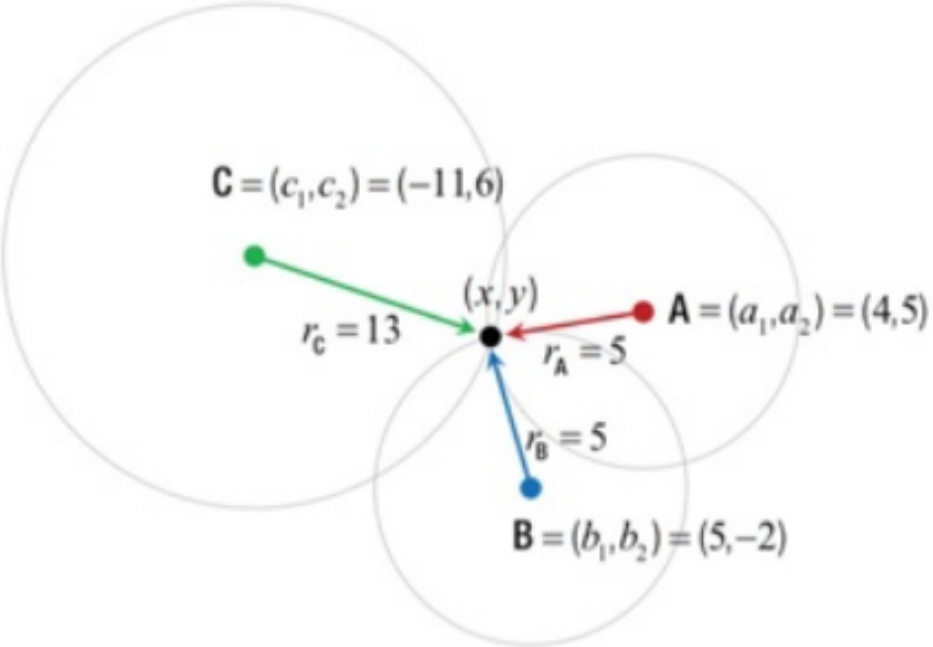
We will not only do this on paper. We will develop algorithms around it to understand the concepts fully. All assignments will contain programming parts.

Motivation

Why learn Numerical Linear Algebra

- Foundation of scientific computations
- Many problems ultimately reduce to linear algebra concepts or algorithms (analytical, computational)
- Examples:
  - Basis of numerical modeling and simulations (FEM, FVM, FD, etc.)
  - Linear Regression, Data fitting, ...
  - PageRank (Google)
- Focus: Fundamental concepts, algorithms, and programming

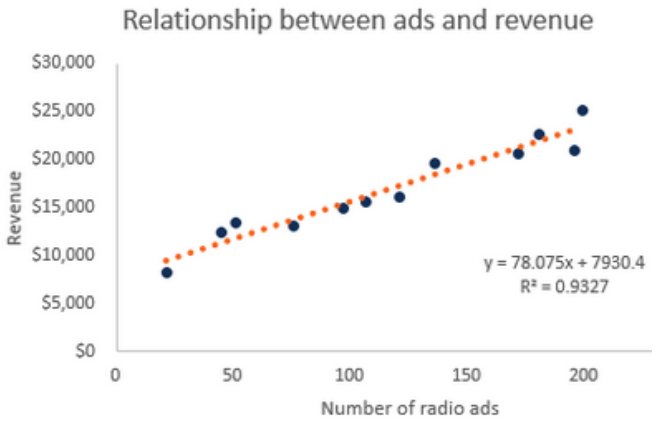
GPS Localization



Linear Regression

Method #3: Simple Linear Regression

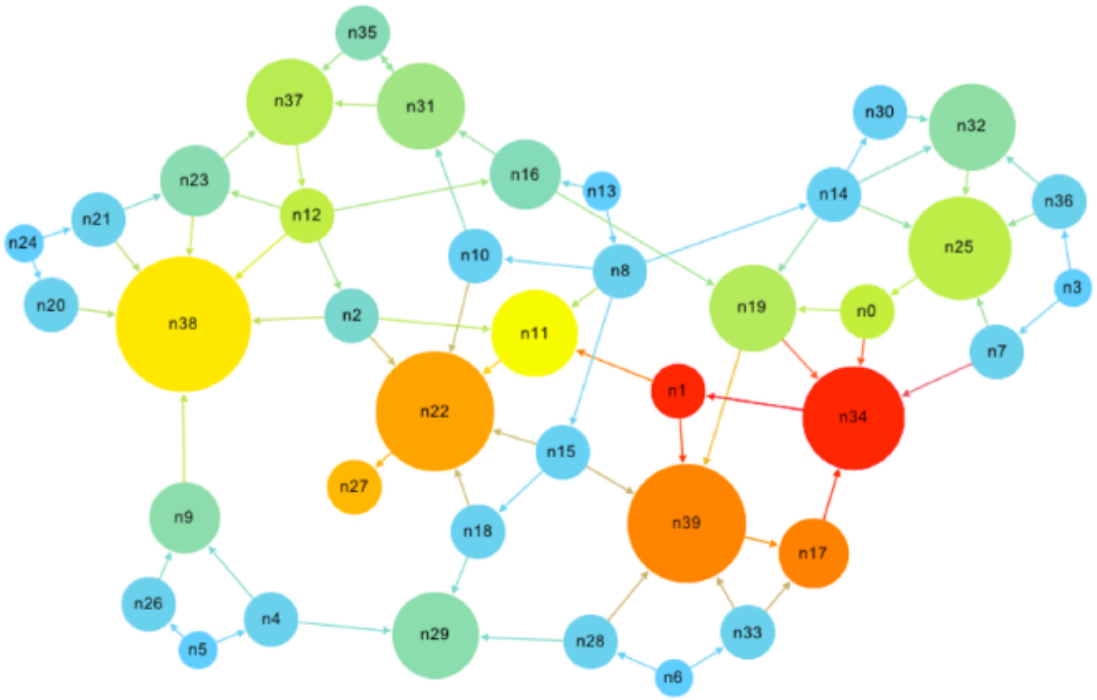
Data	Radio ads	Revenue
Jan	21	$8,350.0
Feb	180	$22,755.0
Mar	50	$13,455.0
Apr	195	$21,100.0
May	96	$15,000.0
Jun	44	$12,500.0
Jul	171	$20,700.0
Aug	135	$19,722.0
Sep	120	$16,115.0
Oct	75	$13,100.0
Nov	106	$15,670.0
Dec	198	$25,300.0
Totals	**1,391**	**$203,767.0**
Average	**116**	**$16,980.6**



Forecast function

100	=FORECAST(B20,$C$4:$C$15,$B$4:$B$15)
150	FORECAST(x, known_ys, known_xs)
200	$23,545.4

Page Rank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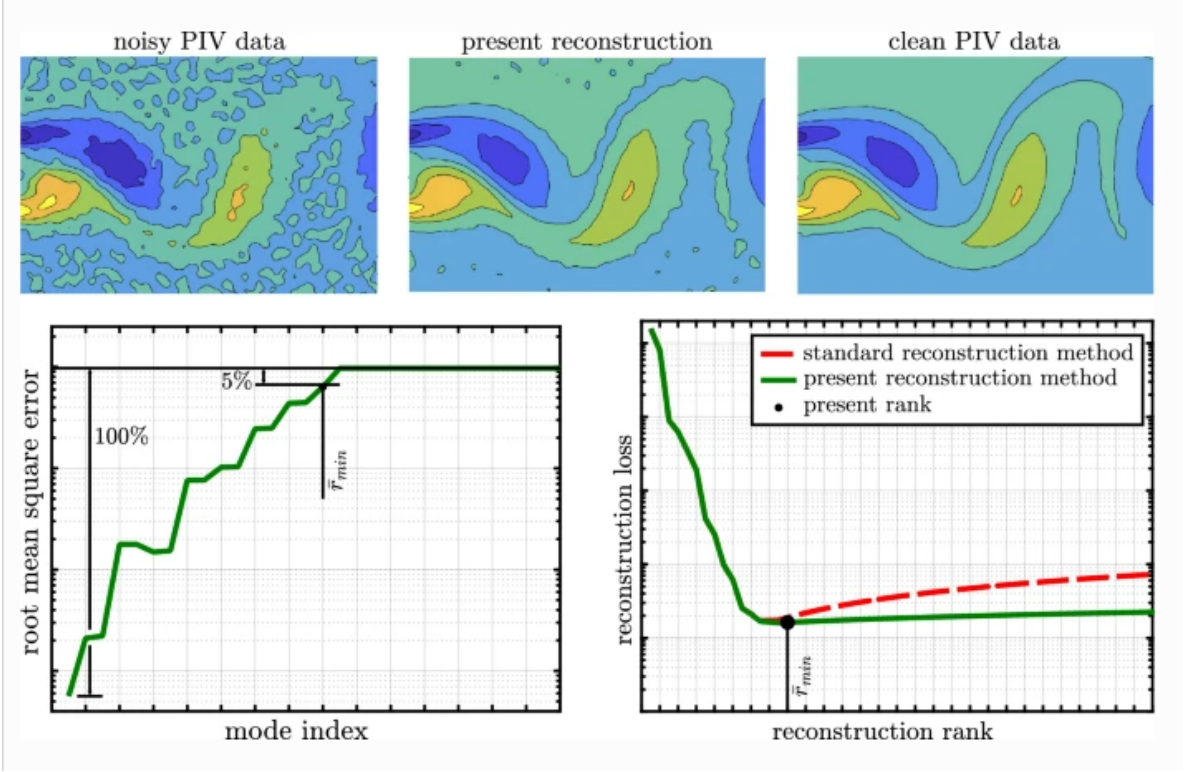
Noise Filtering



Matrix-Vector Multiplication

Definition

- Matrix-vector product $\mathbf{b} = \mathbf{A}\mathbf{x}$

$$b_i = \sum_{j=1}^{n} a_{ij} x_j$$

- All entries belong to $\mathbb{C}$, the field of complex numbers. The space of $m-$vectors is $\mathbb{C}^m$, and the space of $m \times n$ matrices is $\mathbb{C}^{m \times n}$.

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- The map $\mathbf{x} \to \mathbf{A}\mathbf{x}$ is linear, which means for any $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$ and any $\alpha \in \mathbb{C}$

$$\mathbf{A}(\mathbf{x} + \mathbf{y}) = \mathbf{A}\mathbf{x} + \mathbf{A}\mathbf{y}$$

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- In lecture notes, I use **boldface UPPERCASE for matrices, and boldface lowercase letters for vectors.**

Pseudo code for Matrix-Vector Product

This should be straightforward to convert into real computer codes in any programming language.

Pseudo-code for $\mathbf{b} = \mathbf{A}\mathbf{x}$

```
for i = 1 to m do
    b(i) = 0;
    for j = 1 to n do
        b(i) = b(i) + A(i,j) * x(j);
    end for
end for
```

Linear Combination

- Alternatively, matrix-vector product can be viewed as

$$\mathbf{b} = \mathbf{A}\mathbf{x} = \sum_{j=i}^{n} x_j \mathbf{a}_j$$

i.e **b** is a linear combination of column vectors of **A**

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- Two different views of matrix-vector products:
  1. $b_i = \sum_{j=1}^{n} a_{ij} x_j$: **A** acts on **x** to produce **b**; scalar operations
  2. $\mathbf{b} = \sum_{j=i}^{n} x_j \mathbf{a}_j$: **x** acts on **A** to produce **b**; vector operations

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- If **A** is $m \times n$, **Ax** can be viewed as a mapping from $\mathbb{C}^n$ to $\mathbb{C}^m$

Takeaways

- Space travel: Matrix Vector multiplication takes you from one 'space' to another.
- Two ways to do the same thing
  - We like the product expressed as linear combinations
  - Let's convince ourselves: Algorithmic investigation
- Getting started with Python and numpy package/library.